FLEET TEAM DIGEST

737MAX-FTD-34-24009

Issue Title : Unexpected FMC Behavior During Takeoff Phase of Flight While Leveling-Off Below 1500ft

Airplane Model	ATA	Minor Model(s)
737MAX	3461-00	-10, -7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
12/12/2024	03/27/2026	01/16/2026

Estimated Completion Date	Next Update Date
01/31/2028	09/30/2026

Revision Description

Updated the following section(s):

- Milestones

Description

Boeing has received operator reports of unexpected FMC behavior during level-off on a Standard Instrument Departure (SID) leaving Naha Airport. Level-offs are not part of the SID procedure, however, ATC orders level-offs at 1200 feet when there is traffic from a nearby airport. Restrictions are not given in flight charts or navigation database.

Following the level-off at 1200 feet, the FMC behaviors can occur:

- FMC messages appear on the CDU scratchpad:
  - RW/APP TUNE DISAGREE
  - APPRCH VREF NOT SELECTED
- VNAV CLB and CRZ pages are blank
- Lateral Deviation from SID flight path where the plane turns tracking an arc

In all cases, the CLB and CRZ pages returned to normal after entering the cruise target altitude. FMC target speed was activated when VNAV was selected and the flights continued without incident.

Boeing notes that this reported case will cause a transition to the descent flight phase when all of the following conditions are met:

1. The flight phase is takeoff

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- Cases reported to Boeing have occurred during departure from Naha Airport.

2. An edit or a prediction update occurs
   - Every 15 seconds, a trial prediction is performed. Out-of-tolerance prediction will be updated.
3. The airplane is not in takeoff thrust limit mode
   - The takeoff thrust limit mode transitions to cruise thrust limit mode upon transition to altitude hold.
4. The airplane is within 1500' altitude of the published elevation of the departure airport.
   - Level-off at 1200ft to comply with ATC direction.
5. The airplane is above MCP altitude
   - In cases reported to Boeing, the MCP altitude was set to 1200, and an altitude overshoot occurred.
6. The airplane is descending (VS < 0)
   - In cases reviewed by Boeing, the airplane descended back to 1200 following the altitude overshoot.

From the analysis, the scratchpad messages USING RSV FUEL and APPRCH VREF NOT SELECTED appeared due to the transition to descent phase. The FMC considered itself to be in an approach environment, which resulted in the approach speed and frequency not matching the origin values.

The CRZ and CLB pages are blank when airplane is in descent flight phase, and the DES page is active.

Lateral deviation may occur due to the flight path being re-computed when the FMC enters the descent phase. During the re-prediction, any waypoints prior to Top of Descent, are incorrectly considered cruise waypoints and assigned cruise speeds. The vertical path will be built backwards from the destination runway until the entered cruise altitude is reached. The FMC may compute larger turn radii for waypoints due to the incorrect high cruise speeds assigned to the waypoints. This can result in an unexpected turn if LNAV is engaged, due to the FMC following the recalculated flight plan with a larger turn radius. This condition may occur with FMS OPS U10.8A, U11, U12, U13, U14, and U14.1.

While Boeing has only received reports of this behavior occurring at Naha Airport, reported condition can occur on any takeoff as long as the logic for the FMC to transition to descent phase is satisfied.

Status

This condition has been reproduced and its root cause has been determined. Boeing currently plans to incorporate a VNAV Software logic improvement to address this condition in FMC OPS U14.2.

Boeing has determined this condition is Not a Safety Issue by our Continued Operational Safety Process (COSP).

Interim Action

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There are no maintenance tasks to perform since the current software is working as implemented; however, a software improvement will be added to prevent the annunciation of the reported condition. Following a suspected occurrence of this issue, QAR data may be downloaded and sent to Boeing and GE Aviation for confirmation.

To mitigate the possibility of producing the described FMC behaviors, avoid overshooting the altitude set on the MCP and avoid descending maneuvers when the aircraft is within 1500' altitude of the published elevation of the departure airport

Final Action

Incorporate FMS software update U14.2 via Service bulletin when available.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

FMC OPS U14.2 - 737 MAX - Certification | No earlier than 2028 (Target)

FMC OPS U14.2 - 737 MAX - Production Incorporation | No earlier than 2028 (Target)

FMC OPS U14.2 - 737 MAX - Service Bulletin Release | No earlier than 2028 (Target)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-34-23005		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Flight Operations : Flight Operations
- New Equipment : Information on new parts available now or soon to be introduced

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